

Answer Key: Understanding OLS

1. What are we trying to learn?

β is the true but unknown population parameter describing the population relationship between $\mathbf{X}$ and $\mathbf{y}$. The error $\mathbf{u}$ contains omitted and unobserved determinants of $\mathbf{y}$, including random shocks.

2. How is the OLS estimator defined?

β is the true population parameter; $\mathbf{b}$ is an arbitrary candidate coefficient vector; and $\hat{\beta}$ is the candidate that minimizes the sum of squared residuals. OLS chooses the line whose squared prediction errors are smallest.

3. What are OLS residuals?

$\hat{u}_i = y_i - \hat{y}_i$. It is the vertical difference between an observed outcome and its fitted value. It is positive above the fitted line and negative below it.

4. A mechanical property of OLS

$\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$ means the fitted residuals are orthogonal to every included regressor. This is a mechanical property of OLS, not a population assumption.

5. Why is $\hat{\beta}$ different from β ?

Different samples contain different realizations of the unobserved errors, so they produce different estimates. The equation shows that estimation error depends on the regressor variation and on how the regressors happen to align with the errors in the sample.

6. What is exogeneity?

$E(\mathbf{u} | \mathbf{X}) = \mathbf{0}$ means that, conditional on the observed regressors, the unobserved determinants of the outcome have mean zero. It can fail when an omitted factor affects both a regressor and the outcome; for example, unobserved ability may affect both education and wages.

7. Sample residuals versus population errors

$\mathbf{X}'\hat{\mathbf{u}} = \mathbf{0}$ concerns fitted sample residuals and is automatic. $E(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$ concerns the unobserved population error and is an identifying assumption requiring substantive justification.

8. What does unbiasedness mean?

Across repeated samples with the same $\mathbf{X}$, the average OLS estimate equals the true parameter. It does not mean that every sample estimate equals the truth; individual estimates may be above or below it.

9. What determines the precision of OLS?

The estimate is the coefficient calculated from the sample; its standard error measures its sampling variability. A small standard error means greater precision. Precision generally improves with a larger sample, more independent regressor variation, and less error variation. Multicollinearity reduces precision.

10. Why do we estimate the error variance?

σ^2 is unknown because the population errors are unobserved. Residuals approximate those errors. Estimating p coefficients uses p degrees of freedom, leaving $n - p$ independent residual directions; this is why $\hat{\sigma}^2 = SSR/(n - p)$.

11. Why does the normal distribution appear?

Approximate normality lets us quantify how far an estimate is likely to be from the population parameter. It provides the basis for hypothesis tests, p -values, and confidence intervals.

12. Statistical significance

A hypothesis test asks whether the estimate is sufficiently inconsistent with a null value, given its sampling uncertainty. A confidence interval gives a range of parameter values consistent with the data. Statistical significance does not imply economic importance: a tiny effect can be precisely estimated in a large sample.

13. Statistical inference versus causal inference

Statistical inference asks how precisely the population coefficient has been estimated. Causal inference asks what would happen to Y if we intervened and changed X . A biased estimate can have a small standard error, and a noncausal association can be statistically significant. Precision does not remove confounding or endogeneity.

14. What does Gauss–Markov tell us?

“Unbiased” means the estimator equals the true parameter on average. “Best” means lowest variance among linear unbiased estimators under the Gauss–Markov assumptions. The theorem does not establish causality and does not compare OLS with every imaginable estimator.

Final summary

1. OLS 2. sampling uncertainty 3. exogeneity 4. unbiasedness 5. statistical inference.